Chapter 2. Cauchy's Theorem and Its Applications

Goursat's theorem, Cauchy's theorem in a disc, Cauchy integral formulas, Morera, Schwarz reflection, Runge

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Overview

Loosely, Cauchy's theorem says that if f is holomorphic in an open set Ω and $\gamma \subset \Omega$ is a closed curve whose interior is also in Ω, then

$$\int_\gamma f(z)\,dz = 0. \tag{1}$$

A precise general formulation requires defining the "interior" of a curve unambiguously, which is not easy; at this stage we restrict to regions bounded by "toy contours", whose interiors are obvious. The starting point: it suffices that f have a primitive (Corollary 3.3), and primitives exist by *Goursat's theorem*: $\int_T f(z)\,dz = 0$ for a triangle T whose interior lies in Ω. This leads to the *Cauchy integral formula*

$$f(z) = \frac{1}{2\pi i}\int_C \frac{f(\zeta)}{\zeta - z}\,d\zeta \qquad (z \text{ inside } C),$$

and hence to the regularity of holomorphic functions (existence of derivatives of all orders—decisively false for real variables), power series expansions, analytic continuation, Liouville's theorem and the fundamental theorem of algebra, and Morera's theorem.

1 Goursat's theorem

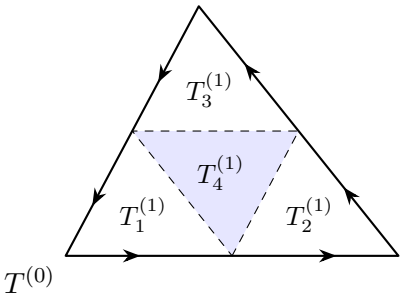


Figure. *Goursat's subdivision: joining the midpoints splits $T^{(0)}$ into four triangles, each oriented like $T^{(0)}$. The dashed interior edges are traversed twice in opposite directions and cancel, so $\int_{\partial T^{(0)}} f = \sum_{j=1}^{4} \int_{\partial T_j^{(1)}} f$, and one of the four integrals is at least a quarter of the total in absolute value.*

By Corollary 3.3, if f has a primitive in Ω then $\int_\gamma f = 0$ for closed $\gamma \subset \Omega$. Conversely, if the integral vanishes for certain curves, a primitive will exist. Goursat's theorem is the starting point from which most other results of this chapter are deduced. (Its interest is that its proof needs only the existence of f' at each point, not its continuity.)

Theorem 1.1

If Ω is an open set in $\mathbb{C}$, and $T \subset \Omega$ a triangle whose interior is also contained in Ω, then

$$\int_T f(z)\,dz = 0$$

whenever f is holomorphic in Ω.

Proof. Let $T^{(0)}$ be the original triangle (positively oriented), with diameter $d^{(0)}$ and perimeter $p^{(0)}$. Bisect each side and connect the midpoints, creating four similar triangles $T_1^{(1)}, \dots, T_4^{(1)}$, oriented consistently with $T^{(0)}$. After cancellation of the integrals over interior sides traversed in opposite directions,

$$\int_{T^{(0)}} f(z)\,dz = \int_{T_1^{(1)}} f + \int_{T_2^{(1)}} f + \int_{T_3^{(1)}} f + \int_{T_4^{(1)}} f. \tag{2}$$

For some j, $\left|\int_{T^{(0)}} f\right| \le 4\left|\int_{T_j^{(1)}} f\right|$, for otherwise (2) would be contradicted. Call this triangle $T^{(1)}$; its diameter and perimeter are $d^{(1)} = \frac{1}{2}d^{(0)}$, $p^{(1)} = \frac{1}{2}p^{(0)}$. Repeating, we obtain triangles $T^{(0)}, T^{(1)}, \dots, T^{(n)}, \dots$ with

$$\left|\int_{T^{(0)}} f(z)\,dz\right| \le 4^n \left|\int_{T^{(n)}} f(z)\,dz\right|, \qquad d^{(n)} = 2^{-n}d^{(0)}, \quad p^{(n)} = 2^{-n}p^{(0)}.$$

Let $\mathcal{T}^{(n)}$ be the solid closed triangle with boundary $T^{(n)}$. These form a nested sequence of compact sets with diameters tending to 0, so by Proposition 1.4 there is a unique point z_0 in all of them. Since f is holomorphic at z_0,

$$f(z) = f(z_0) + f'(z_0)(z - z_0) + \psi(z)(z - z_0), \qquad \psi(z) \to 0 \text{ as } z \to z_0.$$

The constant $f(z_0)$ and the linear function $f'(z_0)(z - z_0)$ have primitives, so by Corollary 3.3

$$\int_{T^{(n)}} f(z)\,dz = \int_{T^{(n)}} \psi(z)(z - z_0)\,dz. \tag{3}$$

z_0 is in the solid triangle $\mathcal{T}^{(n)}$ and z on its boundary, so $|z - z_0| \le d^{(n)}$, and by Proposition 3.1(iii)

$$\left|\int_{T^{(n)}} f(z)\,dz\right| \le \epsilon_n d^{(n)} p^{(n)} = \epsilon_n 4^{-n} d^{(0)} p^{(0)}, \qquad \epsilon_n = \sup_{z \in T^{(n)}} |\psi(z)| \to 0.$$

Hence $\left|\int_{T^{(0)}} f\right| \le 4^n \left|\int_{T^{(n)}} f\right| \le \epsilon_n d^{(0)} p^{(0)}$, and letting $n \to \infty$ concludes the proof. ■

Corollary 1.2

If f is holomorphic in an open set Ω that contains a rectangle R and its interior, then $\int_R f(z)\,dz = 0$.

Proof. Split R by a diagonal into two triangles T_1, T_2, oriented consistently; then $\int_R f = \int_{T_1} f + \int_{T_2} f = 0$. ■

2 Local existence of primitives and Cauchy's theorem in a disc

Motivation. Goursat's theorem handles triangles. To pass to arbitrary closed curves in a disc we construct a primitive F with $F' = f$ by integrating along polygonal paths; then $\int_\gamma f = F(\text{end}) - F(\text{start}) = 0$ for every closed γ.

Theorem 2.1

A holomorphic function in an open disc has a primitive in that disc.

Proof. After a translation we may assume the disc D is centered at 0. For $z \in D$ let γ_z be the polygonal line (at most two segments) from 0 horizontally to $\tilde{z} = \mathrm{Re}(z)$ and then vertically to z, oriented from 0 to z, and define

$$F(z) = \int_{\gamma_z} f(w)\,dw.$$

Fix $z \in D$ and h so small that $z + h \in D$. In $F(z + h) - F(z) = \int_{\gamma_{z+h}} f - \int_{\gamma_z} f$, the integrals over the common segment from the origin cancel; completing a square and a triangle and applying Goursat's theorem for triangles and rectangles, we are left with

$$F(z + h) - F(z) = \int_\eta f(w)\,dw,$$

η the segment from z to $z+h$. Since f is continuous at z, $f(w) = f(z) + \psi(w)$ with $\psi(w) \to 0$ as $w \to z$. Hence

$$F(z+h) - F(z) = f(z) \int_\eta dw + \int_\eta \psi(w)\, dw. \tag{4}$$

The constant 1 has primitive w, so the first integral is h (Theorem 3.2); and $\left| \int_\eta \psi(w)\, dw \right| \le \sup_{w\in\eta} |\psi(w)|\, |h|$. The supremum tends to 0 as $h \to 0$, so (4) gives $\lim_{h\to 0} \frac{F(z+h)-F(z)}{h} = f(z)$. ■

So every holomorphic function has a primitive *locally*. The theorem holds not only for discs but for other sets as well ("toy contours" below).

Theorem 2.2 (Cauchy's theorem for a disc)

If f is holomorphic in a disc, then $\int_\gamma f(z)\, dz = 0$ for any closed curve γ in that disc.

Proof. f has a primitive (Theorem 2.1); apply Corollary 3.3. ■

Corollary 2.3

Suppose f is holomorphic in an open set containing the circle C and its interior. Then $\int_C f(z)\, dz = 0$.

Proof. Let D be the disc bounded by C. There is a slightly larger disc $D' \supset \overline{D}$ on which f is holomorphic; apply Cauchy's theorem in D'. ■

Definition (toy contour)

A **toy contour** is any closed curve whose interior is obvious and for which a construction as in Theorem 2.1 is possible in a neighborhood of the curve and its interior. Its **positive orientation** is the one for which the interior is to the left as we travel along it.

Circles, triangles and rectangles are toy contours. Another important one is the **keyhole** Γ: two almost complete circles, one large and one small, connected by a narrow corridor. If f is holomorphic in a neighborhood of Γ and its interior, it is holomorphic inside a slightly larger keyhole Γ' whose interior contains $\overline{\operatorname{int}\Gamma}$. Fix $z_0 \in \operatorname{int}\Gamma'$; for $z \in \operatorname{int}\Gamma'$ let γ_z be a curve in $\operatorname{int}\Gamma'$ from z_0 to z made of finitely many horizontal or vertical segments. For any other such curve $\tilde{\gamma}_z$, Corollary 1.2 gives $\int_{\gamma_z} f = \int_{\tilde{\gamma}_z} f$, so $F(z) = \int_{\gamma_z} f$ is well defined, and as above F is a primitive of f; hence $\int_\Gamma f = 0$. In general, for a toy contour γ,

$$\int_\gamma f(z)\, dz = 0$$

whenever f is holomorphic in an open set containing γ and its interior. Further examples used later: the rectangular keyhole, the multiple keyhole, the semicircle, the indented semicircle, the sector and the parallelogram. For general curves, Appendix B proves the Jordan curve theorem for piecewise-smooth curves: a simple closed piecewise-smooth curve has a well-defined "simply connected" interior, and Cauchy's theorem holds in that setting.

3 Evaluation of some integrals

We show how some integrals may be evaluated using Cauchy's theorem; a systematic approach via residues is in Chapter 3.

Example 1

If $\xi \in \mathbb{R}$, then

$$e^{-\pi\xi^2} = \int_{-\infty}^{\infty} e^{-\pi x^2} e^{-2\pi i x\xi}\, dx, \tag{5}$$

i.e. $e^{-\pi x^2}$ is its own Fourier transform. For $\xi = 0$ this is the known integral $1 = \int_{-\infty}^{\infty} e^{-\pi x^2}\, dx$. Suppose $\xi > 0$ and let $f(z) = e^{-\pi z^2}$, entire. Let γ_R be the rectangle with vertices R, $R + i\xi$, $-R + i\xi$, $-R$, positively oriented. By Cauchy's theorem

$$\int_{\gamma_R} f(z)\, dz = 0. \tag{6}$$

The integral over the real segment is $\int_{-R}^{R} e^{-\pi x^2}\, dx \to 1$. The integral over the right vertical side is $I(R) = \int_0^{\xi} f(R+iy)\, i\, dy = \int_0^{\xi} e^{-\pi(R^2+2iRy-y^2)}\, i\, dy$, and $|I(R)| \le Ce^{-\pi R^2} \to 0$ (ξ fixed); similarly for the left side. The top side contributes

$$\int_R^{-R} e^{-\pi(x+i\xi)^2}\, dx = -e^{\pi\xi^2} \int_{-R}^{R} e^{-\pi x^2} e^{-2\pi i x\xi}\, dx.$$

Letting $R \to \infty$, (6) gives $0 = 1 - e^{\pi\xi^2} \int_{-\infty}^{\infty} e^{-\pi x^2} e^{-2\pi i x\xi}\, dx$. If $\xi < 0$ use the symmetric rectangle in the lower half-plane. (This *shifting of the contour* has many applications.)

Example 2

$$\int_0^{\infty} \frac{1 - \cos x}{x^2}\, dx = \frac{\pi}{2}.$$

Let $f(z) = (1 - e^{iz})/z^2$ and integrate over the indented semicircle in the upper half-plane: the segments $[-R, -\epsilon]$ and $[\epsilon, R]$, the small semicircle γ_ϵ^+ of radius ϵ (negatively oriented) and the large one γ_R^+ (positively oriented). By Cauchy's theorem

$$\int_{-R}^{-\epsilon} \frac{1 - e^{ix}}{x^2}\, dx + \int_{\gamma_\epsilon^+} \frac{1 - e^{iz}}{z^2}\, dz + \int_{\epsilon}^{R} \frac{1 - e^{ix}}{x^2}\, dx + \int_{\gamma_R^+} \frac{1 - e^{iz}}{z^2}\, dz = 0.$$

Since $|(1-e^{iz})/z^2| \le 2/|z|^2$ in the upper half-plane, the integral over γ_R^+ tends to 0 as $R \to \infty$, so $\int_{|x|\ge\epsilon} \frac{1-e^{ix}}{x^2}\, dx = -\int_{\gamma_\epsilon^+} \frac{1-e^{iz}}{z^2}\, dz$. Now $f(z) = \frac{-iz}{z^2} + E(z)$ with E bounded near 0, and on γ_ϵ^+, $z = \epsilon e^{i\theta}$, $dz = i\epsilon e^{i\theta} d\theta$, so

$$\int_{\gamma_\epsilon^+} \frac{1 - e^{iz}}{z^2}\, dz \to \int_{\pi}^{0} (-i)(i)\, d\theta = -\pi \quad \text{as } \epsilon \to 0.$$

Taking real parts, $\int_{-\infty}^{\infty} \frac{1-\cos x}{x^2}\, dx = \pi$; the integrand is even.

4 Cauchy's integral formulas

Motivation. Cauchy's theorem says integrals of holomorphic functions over closed curves vanish; applied to $f(\zeta)/(\zeta - z)$ it says much more, that the values of f inside a disc are determined by its values on the boundary circle. Differentiating under the integral then shows that holomorphic functions are infinitely differentiable and equal to their Taylor series, which is the key difference from real analysis.

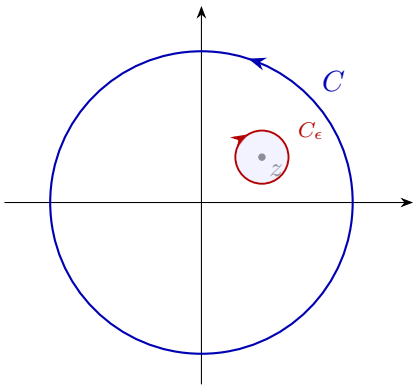


Figure. *For z inside C, $\zeta \mapsto f(\zeta)/(\zeta - z)$ is holomorphic between C and a small circle C_ϵ around z, so both circles give the same integral. As $\epsilon \to 0$ the small integral tends to $2\pi i f(z)$, giving $f(z) = \frac{1}{2\pi i}\int_C \frac{f(\zeta)}{\zeta - z}\,d\zeta$.*

Integral representations recover a function on a large set from its values on a smaller set (cf. the Poisson integral formula for the steady-state heat equation in the disc).

Theorem 4.1

Suppose f is holomorphic in an open set containing the closure of a disc D. If C denotes the boundary circle of this disc with the positive orientation, then

$$f(z) = \frac{1}{2\pi i}\int_C \frac{f(\zeta)}{\zeta - z}\,d\zeta \qquad \text{for any point } z \in D.$$

Proof. Fix $z \in D$ and consider the keyhole $\Gamma_{\delta,\epsilon}$ omitting z: the circle C and a small circle of radius ϵ about z, joined by a corridor of width δ. $F(\zeta) = f(\zeta)/(\zeta - z)$ is holomorphic away from $\zeta = z$, so $\int_{\Gamma_{\delta,\epsilon}} F(\zeta)\,d\zeta = 0$ by Cauchy's theorem for this toy contour. Letting $\delta \to 0$, continuity of F makes the integrals over the two sides of the corridor cancel, leaving C (positively oriented) and the small circle C_ϵ about z (negatively oriented). Write

$$F(\zeta) = \frac{f(\zeta) - f(z)}{\zeta - z} + \frac{f(z)}{\zeta - z}. \tag{8}$$

The first term is bounded (as f is holomorphic), so its integral over C_ϵ tends to 0 as $\epsilon \to 0$. And

$$\int_{C_\epsilon} \frac{f(z)}{\zeta - z}\,d\zeta = -f(z)\int_0^{2\pi} \frac{\epsilon i e^{-it}}{\epsilon e^{-it}}\,dt = -f(z)2\pi i.$$

So in the limit $0 = \int_C \frac{f(\zeta)}{\zeta - z}\,d\zeta - 2\pi i f(z)$. ∎

Remark

The same holds for other toy contours: if f is holomorphic in an open set containing a positively oriented rectangle R and its interior, then $f(z) = \frac{1}{2\pi i}\int_R \frac{f(\zeta)}{\zeta - z}\,d\zeta$ for z in the interior of R (use a rectangular keyhole). The integral vanishes for z outside R, since then $f(\zeta)/(\zeta - z)$ is holomorphic inside R.

Corollary 4.2

If f is holomorphic in an open set Ω, then f has infinitely many complex derivatives in Ω.

Moreover, if $C \subset \Omega$ is a circle whose interior is also contained in Ω, then

$$f^{(n)}(z) = \frac{n!}{2\pi i} \int_C \frac{f(\zeta)}{(\zeta - z)^{n+1}}\, d\zeta$$

for all z in the interior of C.

Proof. Induction on n; $n = 0$ is the Cauchy integral formula. Suppose f has up to $n-1$ complex derivatives with $f^{(n-1)}(z) = \frac{(n-1)!}{2\pi i} \int_C \frac{f(\zeta)}{(\zeta - z)^n}\, d\zeta$. For small h,

$$\frac{f^{(n-1)}(z+h) - f^{(n-1)}(z)}{h} = \frac{(n-1)!}{2\pi i} \int_C f(\zeta) \frac{1}{h} \Big[\frac{1}{(\zeta - z - h)^n} - \frac{1}{(\zeta - z)^n} \Big] d\zeta. \tag{9}$$

Since $A^n - B^n = (A-B)[A^{n-1} + A^{n-2}B + \cdots + AB^{n-2} + B^{n-1}]$, with $A = 1/(\zeta - z - h)$, $B = 1/(\zeta - z)$ the bracket in (9) is

$$\frac{h}{(\zeta - z - h)(\zeta - z)} [A^{n-1} + A^{n-2}B + \cdots + AB^{n-2} + B^{n-1}].$$

For small h, $z + h$ and z stay at a finite distance from C, so as $h \to 0$ the quotient converges to

$$\frac{(n-1)!}{2\pi i} \int_C f(\zeta) \frac{1}{(\zeta - z)^2} \frac{n}{(\zeta - z)^{n-1}}\, d\zeta = \frac{n!}{2\pi i} \int_C \frac{f(\zeta)}{(\zeta - z)^{n+1}}\, d\zeta.$$

■

The formulas of Theorem 4.1 and Corollary 4.2 are the **Cauchy integral formulas**.

Corollary 4.3 (Cauchy inequalities)

If f is holomorphic in an open set that contains the closure of a disc D centered at z_0 and of radius R, then

$$|f^{(n)}(z_0)| \le \frac{n!\, \|f\|_C}{R^n},$$

where $\|f\|_C = \sup_{z \in C} |f(z)|$ denotes the supremum of $|f|$ on the boundary circle C.

Proof. By the Cauchy integral formula,

$$|f^{(n)}(z_0)| = \Big| \frac{n!}{2\pi i} \int_C \frac{f(\zeta)}{(\zeta - z_0)^{n+1}}\, d\zeta \Big| = \frac{n!}{2\pi} \Big| \int_0^{2\pi} \frac{f(z_0 + Re^{i\theta})}{(Re^{i\theta})^{n+1}} Rie^{i\theta}\, d\theta \Big| \le \frac{n!}{2\pi} \frac{\|f\|_C}{R^n} 2\pi.$$

■

Theorem 4.4

Suppose f is holomorphic in an open set Ω. If D is a disc centered at z_0 and whose closure is contained in Ω, then f has a power series expansion at z_0

$$f(z) = \sum_{n=0}^{\infty} a_n (z - z_0)^n$$

for all $z \in D$, and the coefficients are given by $a_n = \dfrac{f^{(n)}(z_0)}{n!}$ for all $n \ge 0$.

Proof. Fix $z \in D$. By the Cauchy integral formula

$$f(z) = \frac{1}{2\pi i} \int_C \frac{f(\zeta)}{\zeta - z}\, d\zeta, \tag{10}$$

C the boundary of D. Write

$$\frac{1}{\zeta - z} = \frac{1}{\zeta - z_0 - (z - z_0)} = \frac{1}{\zeta - z_0} \cdot \frac{1}{1 - \left(\frac{z - z_0}{\zeta - z_0}\right)}. \tag{11}$$

Since $\zeta \in C$ and $z \in D$ is fixed, there is $0 < r < 1$ with $\left|\frac{z-z_0}{\zeta - z_0}\right| < r$, so

$$\frac{1}{1 - \left(\frac{z - z_0}{\zeta - z_0}\right)} = \sum_{n=0}^{\infty} \left(\frac{z - z_0}{\zeta - z_0}\right)^n, \tag{12}$$

uniformly for $\zeta \in C$. Interchanging sum and integral in (10)–(12),

$$f(z) = \sum_{n=0}^{\infty} \left(\frac{1}{2\pi i} \int_C \frac{f(\zeta)}{(\zeta - z_0)^{n+1}}\, d\zeta\right)(z - z_0)^n.$$

The formula for a_n follows from the Cauchy integral formulas for derivatives (or by differentiating the series). ■

This gives another proof that holomorphic functions are indefinitely differentiable. The expansion at z_0 converges in *any* disc centered at z_0 whose closure lies in Ω; in particular an entire function has a power series expansion about 0 converging on all of $\mathbb{C}$.

Corollary 4.5 (Liouville's theorem)

If f is entire and bounded, then f is constant.

Proof. It suffices to show $f' = 0$ ($\mathbb{C}$ is connected; Corollary 3.4). For each $z_0 \in \mathbb{C}$ and $R > 0$ the Cauchy inequalities give $|f'(z_0)| \le B/R$, B a bound for f. Let $R \to \infty$. ■

Corollary 4.6

Every non-constant polynomial $P(z) = a_n z^n + \cdots + a_0$ with complex coefficients has a root in $\mathbb{C}$.

Proof. If P has no roots, $1/P$ is a bounded entire function. Indeed, with $a_n \neq 0$, $\frac{P(z)}{z^n} = a_n + \left(\frac{a_{n-1}}{z} + \cdots + \frac{a_0}{z^n}\right)$ for $z \neq 0$; the parenthesis tends to 0 as $|z| \to \infty$, so with $c = |a_n|/2$ there is R with $|P(z)| \ge c|z|^n$ for $|z| > R$. P is continuous with no roots in $|z| \le R$, so it is bounded below there as well. By Liouville's theorem $1/P$ is constant, contradicting that P is non-constant. ■

Corollary 4.7

Every polynomial $P(z) = a_n z^n + \cdots + a_0$ of degree $n \ge 1$ has precisely n roots in $\mathbb{C}$. If these roots are denoted by $w_1, \ldots, w_n$, then P can be factored as $P(z) = a_n(z - w_1)(z - w_2)\cdots(z - w_n)$.

Proof. P has a root w_1. Writing $z = (z - w_1) + w_1$ and expanding, $P(z) = b_n(z - w_1)^n + \cdots + b_1(z - w_1) + b_0$ with $b_n = a_n$; $P(w_1) = 0$ forces $b_0 = 0$, so $P(z) = (z - w_1)Q(z)$ with Q of degree $n - 1$. By induction $P(z) = c(z - w_1)\cdots(z - w_n)$, and comparing coefficients of z^n, $c = a_n$. ■

Finally, *analytic continuation*: the "genetic code" of a holomorphic function is determined by its values on arbitrarily small sets. Here Ω must be connected.

Theorem 4.8

Suppose f is a holomorphic function in a region Ω that vanishes on a sequence of distinct points with a limit point in Ω. Then f is identically 0.

Proof. Let $z_0 \in \Omega$ be a limit point of $\{w_k\}$ with $f(w_k) = 0$. First, $f \equiv 0$ near z_0: on a disc $D \subset \Omega$ centered at z_0, $f(z) = \sum a_n(z - z_0)^n$. If $f \not\equiv 0$ on D, let m be the smallest integer with $a_m \neq 0$; then $f(z) = a_m(z - z_0)^m(1 + g(z - z_0))$ with $g(z - z_0) \to 0$ as $z \to z_0$. Taking $z = w_k \neq z_0$ close to z_0, $a_m(w_k - z_0)^m \neq 0$ and $1 + g(w_k - z_0) \neq 0$, contradicting $f(w_k) = 0$.

Now let U be the interior of $\{z \in \Omega : f(z) = 0\}$; U is open and non-empty. U is closed in Ω: if $z_n \in U$ and $z_n \to z \in \Omega$, then $f(z) = 0$ by continuity, and f vanishes near z by the argument just given, so $z \in U$. With $V = \Omega - U$, U and V are disjoint open sets with $\Omega = U \cup V$; Ω is connected and $z_0 \in U$, so $U = \Omega$. ■

Corollary 4.9

Suppose f and g are holomorphic in a region Ω and $f(z) = g(z)$ for all z in some non-empty open subset of Ω (or more generally for z in some sequence of distinct points with limit point in Ω). Then $f(z) = g(z)$ throughout Ω.

If f, F are analytic in regions $\Omega \subset \Omega'$ and agree on Ω, F is an **analytic continuation** of f into Ω'; by the corollary it is unique.

5 Further applications

Motivation. The integral formula turns many analytic questions into estimates of integrals. We collect the consequences used repeatedly later: a converse of Cauchy's theorem (Morera), the stability of holomorphy under uniform limits and under integration in a parameter, the reflection principle, and approximation by polynomials and rational functions.

5.1 Morera's theorem

Theorem 5.1 (Morera)

Suppose f is a continuous function in the open disc D such that for any triangle T contained in D, $\int_T f(z)\,dz = 0$. Then f is holomorphic.

Proof. By the proof of Theorem 2.1, f has a primitive F in D with $F' = f$. By regularity (Corollary 4.2), F is indefinitely (hence twice) complex differentiable, so f is holomorphic. ■

5.2 Sequences of holomorphic functions

Theorem 5.2

If $\{f_n\}_{n=1}^{\infty}$ is a sequence of holomorphic functions that converges uniformly to a function f in every compact subset of Ω, then f is holomorphic in Ω.

Proof. Let D be a disc with $\overline{D} \subset \Omega$ and T a triangle in D. By Goursat, $\int_T f_n = 0$ for all n. Since $f_n \to f$ uniformly on $\overline{D}$, f is continuous and $\int_T f_n \to \int_T f$, so $\int_T f = 0$, and by Morera's theorem f is holomorphic in D. This holds for every such D. ■

This fails for real variables: by Weierstrass's theorem every continuous function on $[0,1]$ is a uniform limit of polynomials, yet not every continuous function is differentiable.

Theorem 5.3

Under the hypotheses of the previous theorem, the sequence of derivatives $\{f_n'\}_{n=1}^{\infty}$ converges uniformly to f' on every compact subset of Ω.

Proof. We may assume the convergence is uniform on all of Ω. For $\delta > 0$ let $\Omega_\delta = \{z \in \Omega : \overline{D_\delta(z)} \subset \Omega\}$, the points at distance $> \delta$ from the boundary. It suffices to show $f_n' \to f'$ uniformly on each Ω_δ, which follows by applying to $F = f_n - f$ the inequality

$$\sup_{z\in\Omega_\delta} |F'(z)| \le \frac{1}{\delta} \sup_{\zeta\in\Omega} |F(\zeta)|, \tag{13}$$

valid for F holomorphic in Ω. For $z \in \Omega_\delta$, $\overline{D_\delta(z)} \subset \Omega$ and by the Cauchy integral formula

$$|F'(z)| = \Big|\frac{1}{2\pi i}\int_{C_\delta(z)} \frac{F(\zeta)}{(\zeta - z)^2}\, d\zeta\Big| \le \frac{1}{2\pi} \sup_{\zeta\in\Omega}|F(\zeta)| \frac{1}{\delta^2} 2\pi\delta = \frac{1}{\delta}\sup_{\zeta\in\Omega}|F(\zeta)|.$$

■

Likewise $f_n^{(k)} \to f^{(k)}$ uniformly on compact subsets for every $k \ge 0$. In practice Theorem 5.2 constructs holomorphic functions as series $F(z) = \sum_{n=1}^{\infty} f_n(z)$ (14) converging uniformly on compact subsets of Ω, e.g. the zeta function (Chapter 6).

5.3 Holomorphic functions defined in terms of integrals

After a linear change of variables we take the interval $[0,1]$.

Theorem 5.4

Let $F(z,s)$ be defined for $(z,s) \in \Omega \times [0,1]$ where Ω is an open set in $\mathbb{C}$. Suppose F satisfies the following properties:
(i) $F(z,s)$ is holomorphic in z for each s.
(ii) F is continuous on $\Omega \times [0,1]$. Then the function f defined on Ω by $f(z) = \int_0^1 F(z,s)\, ds$ is holomorphic.

Proof. Instead of justifying an interchange of integrals in Morera's theorem, view the integral as a uniform limit of Riemann sums. Let $f_n(z) = \frac{1}{n}\sum_{k=1}^{n} F(z, k/n)$, holomorphic in Ω by (i). On a disc D with $\overline{D} \subset \Omega$, F is uniformly continuous on $\overline{D} \times [0,1]$, so given $\epsilon > 0$ there is $\delta > 0$ with $\sup_{z\in\overline{D}} |F(z,s_1) - F(z,s_2)| < \epsilon$ when $|s_1 - s_2| < \delta$. For $n > 1/\delta$ and $z \in D$,

$$|f_n(z) - f(z)| = \Big|\sum_{k=1}^{n} \int_{(k-1)/n}^{k/n} [F(z,k/n) - F(z,s)]\, ds\Big| \le \sum_{k=1}^{n} \frac{\epsilon}{n} = \epsilon.$$

So $f_n \to f$ uniformly on D, and f is holomorphic in D by Theorem 5.2; hence in Ω. ■

5.4 Schwarz reflection principle

Holomorphic functions are rigid: there are holomorphic functions in a disc, continuous on its closure, that cannot be continued beyond it; and they vanish identically if they vanish on a line segment. Yet there is a simple and useful extension phenomenon. Let Ω be open and symmetric with respect to the real line ($z \in \Omega \iff \overline{z} \in \Omega$), Ω^+ its part in the upper half-plane, Ω^- the part in the lower half-plane, and $I = \Omega \cap \mathbb{R}$; then $\Omega = \Omega^+ \cup I \cup \Omega^-$.

Theorem 5.5 (Symmetry principle)

If f^+ and f^- are holomorphic functions in Ω^+ and Ω^- respectively, that extend continuously to I and $f^+(x) = f^-(x)$ for all $x \in I$, then the function f defined on Ω by

$$f(z) = \begin{cases} f^+(z) & z \in \Omega^+, \\ f^+(z) = f^-(z) & z \in I, \\ f^-(z) & z \in \Omega^- \end{cases}$$

is holomorphic on all of Ω.

Proof. f is continuous on Ω; the issue is holomorphy at points of I. Let $D \subset \Omega$ be a disc centered at a point of I; we apply Morera's theorem. If a triangle $T \subset D$ does not meet I, $\int_T f = 0$ since f is holomorphic in the half-discs. If one side or vertex of T lies on I and the rest is, say, in the upper half-disc, let T_ϵ be obtained by raising that edge or vertex by ϵ; $\int_{T_\epsilon} f = 0$ as T_ϵ lies in the upper half-disc, and letting $\epsilon \to 0$, by continuity $\int_T f = 0$. If the interior of T meets I, write T as a union of triangles each with an edge or vertex on I. By Morera's theorem f is holomorphic in D. ■

Theorem 5.6 (Schwarz reflection principle)

Suppose that f is a holomorphic function in Ω^+ that extends continuously to I and such that f is real-valued on I. Then there exists a function F holomorphic in all of Ω such that $F = f$ on Ω^+.

Proof. Define $F(z) = \overline{f(\overline{z})}$ for $z \in \Omega^-$. If $z, z_0 \in \Omega^-$ then $\overline{z}, \overline{z_0} \in \Omega^+$ and $f(\overline{z}) = \sum a_n(\overline{z} - \overline{z_0})^n$ near $\overline{z_0}$, so $F(z) = \sum \overline{a_n}(z - z_0)^n$ and F is holomorphic in Ω^-. Since f is real on I, $\overline{f(x)} = f(x)$ for $x \in I$, so F extends continuously up to I and agrees with f there. Apply the symmetry principle. ■

5.5 Runge's approximation theorem

By Weierstrass's theorem, continuous functions on a compact interval are uniform limits of polynomials. What conditions on a compact $K \subset \mathbb{C}$ ensure that every function holomorphic in a neighborhood of K is a uniform limit of polynomials on K? For a disc this follows from power series. Some condition is necessary: for $f(z) = 1/z$ on the unit circle $K = C$, $\int_C f = 2\pi i$ while $\int_C p = 0$ for every polynomial p, so uniform approximation is impossible. The right condition is that K^c be connected (a variant of this example shows it is also necessary).

Theorem 5.7 (Runge)

Any function holomorphic in a neighborhood of a compact set K can be approximated uniformly on K by rational functions whose singularities are in K^c. If K^c is connected, any function holomorphic in a neighborhood of K can be approximated uniformly on K by polynomials.

The second part follows from the first by "pushing the singularities to infinity". The key is an integral representation from the Cauchy formula for squares:

Lemma 5.8

Suppose f is holomorphic in an open set Ω, and $K \subset \Omega$ is compact. Then there exist finitely

many segments $\gamma_1, \dots, \gamma_N$ in $\Omega - K$ such that

$$f(z) = \sum_{n=1}^{N} \frac{1}{2\pi i} \int_{\gamma_n} \frac{f(\zeta)}{\zeta - z}\, d\zeta \qquad \text{for all } z \in K. \tag{15}$$

Proof. Let $d = c \cdot d(K, \Omega^c)$ with $c < 1/\sqrt{2}$, and consider a grid of closed squares with sides of length d parallel to the axes. Let $\mathcal{Q} = \{Q_1, \dots, Q_M\}$ be the squares of the grid meeting K, with positively oriented boundaries ∂Q_m, and let $\gamma_1, \dots, \gamma_N$ be the sides of squares in $\mathcal{Q}$ that do not belong to two adjacent squares of $\mathcal{Q}$. By the choice of d, each γ_n lies in Ω and does not meet K (if it did, it would belong to two adjacent squares of $\mathcal{Q}$). For $z \in K$ not on the boundary of a square of $\mathcal{Q}$, $z \in Q_j$ for some j, and by Cauchy's theorem and formula

$$\frac{1}{2\pi i} \int_{\partial Q_m} \frac{f(\zeta)}{\zeta - z}\, d\zeta = \begin{cases} f(z) & m = j, \\ 0 & m \neq j, \end{cases} \qquad \text{so} \qquad f(z) = \sum_{m=1}^{M} \frac{1}{2\pi i} \int_{\partial Q_m} \frac{f(\zeta)}{\zeta - z}\, d\zeta.$$

The integrals over a common side of two adjacent squares are taken once in each direction and cancel; this gives (15) for such z. Since the γ_n lie in K^c, continuity extends (15) to all $z \in K$. ∎

Lemma 5.9

For any line segment γ entirely contained in $\Omega - K$, there exists a sequence of rational functions with singularities on γ that approximate the integral $\int_\gamma f(\zeta)/(\zeta - z)\, d\zeta$ uniformly on K.

Proof. With a parametrization $\gamma(t)$, $t \in [0,1]$, $\int_\gamma \frac{f(\zeta)}{\zeta - z}\, d\zeta = \int_0^1 \frac{f(\gamma(t))\gamma'(t)}{\gamma(t) - z}\, dt$. Since γ does not meet K, the integrand $F(z,t)$ is jointly continuous on the compact set $K \times [0,1]$, hence uniformly continuous, and as in the proof of Theorem 5.4 the Riemann sums of $\int_0^1 F(z,t)\, dt$ approximate it uniformly on K. Each Riemann sum is a rational function with singularities on γ. ∎

Lemmas 5.8 and 5.9 prove the first part of Theorem 5.7. A rational function whose only singularity is z_0 is a polynomial in $1/(z - z_0)$, so the second part follows from:

Lemma 5.10

If K^c is connected and $z_0 \notin K$, then the function $1/(z - z_0)$ can be approximated uniformly on K by polynomials.

Proof. Choose z_1 outside a large open disc D centered at 0 containing K. Then

$$\frac{1}{z - z_1} = -\frac{1}{z_1} \frac{1}{1 - z/z_1} = -\sum_{n=0}^{\infty} \frac{z^n}{z_1^{n+1}},$$

uniformly for $z \in K$; the partial sums are polynomials approximating $1/(z - z_1)$ uniformly on K, and hence any power $1/(z - z_1)^k$ is a uniform limit of polynomials on K. It now suffices to approximate $1/(z - z_0)$ uniformly on K by polynomials in $1/(z - z_1)$. Since K^c is connected, there is a curve $\gamma \subset K^c$, $\gamma(0) = z_0$, $\gamma(1) = z_1$. Let $\rho = \frac{1}{2} d(K, \gamma) > 0$ (both are compact), and choose points $w_0 = z_0, w_1, \dots, w_\ell = z_1$ on γ with $|w_j - w_{j+1}| < \rho$.

Claim: if $w, w' \in \gamma$ with $|w - w'| < \rho$, then $1/(z - w)$ is a uniform limit on K of polynomials in $1/(z - w')$. Indeed

$$\frac{1}{z - w} = \frac{1}{z - w'} \cdot \frac{1}{1 - \frac{w - w'}{z - w'}} = \sum_{n=0}^{\infty} \frac{(w - w')^n}{(z - w')^{n+1}},$$

and the series converges uniformly for $z \in K$ since $|z - w'| \geq d(K, \gamma) = 2\rho$ gives $|w - w'|/|z - w'| < \rho/(2\rho) = 1/2$. Applying the claim successively, $1/(z - w_0)$ is a uniform limit of polynomials in $1/(z - w_1)$, hence (as powers of $1/(z - w_1)$ are limits of polynomials in $1/(z - w_2)$) of polynomials in $1/(z - w_2)$, and so on, up to polynomials in $1/(z - w_\ell) = 1/(z - z_1)$. ∎